

ThermalManhattan: Robust Thermal Compass under Radiometric Flatness in Manhattan World (Supplementary Material)

Hyeonji Shin[✉], Junpyo Kim[✉], and Pyojin Kim^{*✉}

Gwangju Institute of Science and Technology (GIST), South Korea
{[localization,junpyokim](mailto:localization,junpyokim@gm.gist.ac.kr)}@gm.gist.ac.kr
<https://ThermalManhattan.github.io/>

This supplementary material provides additional analyses and qualitative results complementing the main paper. We present robustness studies (Sec. 1), validation of the RTK-anchored pseudo-ground-truth trajectory (Sec. 2), additional dataset details (Sec. 3), and qualitative comparisons of monocular SLAM baselines on thermal imagery (Sec. 4).

1 Robustness of Our Method

1.1 Ablation Study on Line Detectors

Table 1: Robustness to the line source (mean ARE in $^{\circ}$, averaged over Residential-closed/opened and S1 Building; identical gravity prior for all methods). 1-Line RANSAC* collapses on PLNet lines, whereas our MnS search remains stable across line sources.

Method	LSD	PLNet
1-Line RANSAC* [7]	0.659	5.607
Ours	0.608	0.518

1-Line RANSAC* [7]: the calibrated single-line solver of [7], run with the same known gravity prior as all other estimators so that the vertical is fixed and only the yaw is estimated, as in MWMS and ours.

Tab. 1 isolates the estimator behavior by line detector, with the vertical (gravity) direction fixed identically for both methods. On clean LSD lines, our MnS search and 1-Line RANSAC* [7] are comparable (0.608 vs. 0.659). The difference is dramatic on wireframe (PLNet) lines: the single-line solver collapses (5.607 on average; 15.7° on Residential-opened alone), whereas the MnS search stays around 0.518. The MnS estimate is global and memoryless over all lines, so a few spurious wireframe lines cannot dominate the estimate, whereas a single-line hypothesis can be strongly affected by an outlier.

* Corresponding author: pjinkim@gist.ac.kr

1.2 Robustness on Non-Uniformity Correction (NUC)

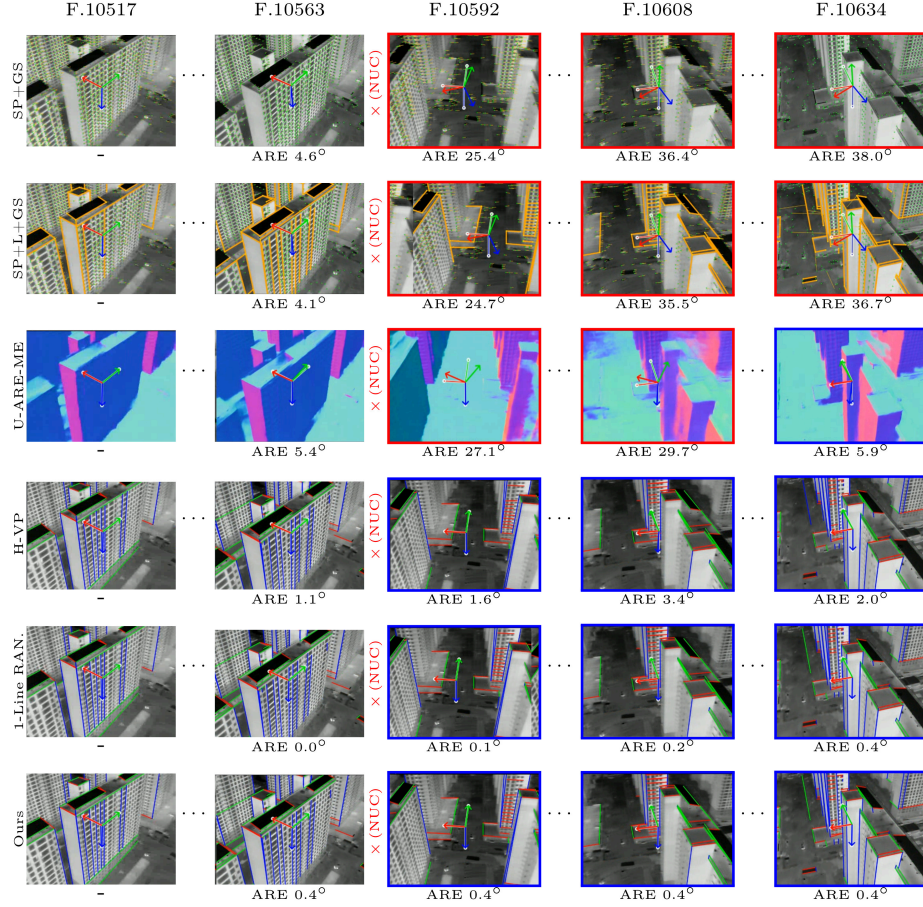


Fig. 1: Rotation estimates across a Non-Uniformity Correction (NUC) event on Residential-closed. The NUC freezes the sensor stream between frames 10563 and 10592 ($\times$ (NUC)), producing a large viewpoint gap. Per-frame ARE is reported below each image; a dash (-) at frame 10517 marks the reference frame against which the subsequent errors are measured. Red and blue boxes mark erroneous and accurate estimates, respectively.

Thermal sensors periodically perform Non-Uniformity Correction (NUC), briefly freezing the image stream and producing a viewpoint jump. Fig. 1 shows such an event on Residential-closed, where the stream freezes between frames 10563 and 10592. Matching-based estimators diverge across the gap, whereas line-regularity-based estimators respond correctly immediately afterward because

each frame is solved independently. U-ARE-ME exhibits elevated error for tens of frames after the gap, while our estimate remains accurate immediately after the NUC event with no recovery transient.

1.3 Robustness on Sparse Features

Fig. 2 shows a segment of Residential-closed where the viewpoint sweeps past the boundary of the complex, leaving only a few structural lines in view. Matching- and normal-based estimators degrade as correspondences and stable normals vanish, and the sampling-based H-VP and 1-Line RANSAC* swing sharply once an outlier line enters the minimal sample. Our MnS search instead counts the consensus of all lines, so a few spurious lines cannot outvote the estimate and the error stays sub-degree throughout.

2 Validation of the Pseudo Ground-Truth Trajectory

Prior work has repeatedly reported that structure-from-motion (SfM) is unreliable on thermal imagery [2, 5, 6, 10], and we reproduce this on our own data: COLMAP with standard sequential matching yields trajectories that substantially deviate from the RTK-GPS trajectories (Fig. 3a). Obtaining accurate ground truth for outdoor thermal sequences is inherently difficult. Delaune *et al.* [3] rely on motion capture and are therefore restricted to a spatially confined outdoor volume, while Tex-NeRF [10] circumvents the limitations of thermal SfM by fusing RGB images into the reconstruction. Neither strategy scales to our setting of wide-area flights (approximately 150×150 m) in extreme darkness, where RGB imagery is severely degraded and cannot be relied upon consistently. We instead anchor the reconstruction with the onboard RTK-GPS, whose positions serve both to seed image pairing through a spatial matcher and to constrain bundle adjustment (BA) as camera-center position priors.

Fig. 3 isolates the contribution of each component, with all variants evaluated against the RTK-GPS positions. Sequential matching alone (Fig. 3a) produces an arbitrary-scale reconstruction with large trajectory error. Adding the RTK-GPS position priors to BA via `pose_prior_mapper` (Fig. 3b) brings only marginal improvement, because sequential matching still fails to establish sufficient loop-closing constraints for a globally consistent reconstruction. Only when the RTK-GPS additionally seeds *spatial* matching (Fig. 3c) does the reconstruction show clear improvement in agreement with the RTK-GPS trajectory. RTK-GPS-guided spatial matching improves correspondence connectivity, while the position priors provide metric anchoring during BA.

This design improves the reliability of the pseudo-GT 6-DoF poses by incorporating RTK-GPS information into the SfM reconstruction. RTK-GPS anchors the camera positions and guides spatial image matching, providing stronger correspondence connectivity and geometric constraints for bundle adjustment than sequential matching alone. Within this RTK-anchored reconstruction, thermal-image correspondences refine the camera poses, including their orientations,

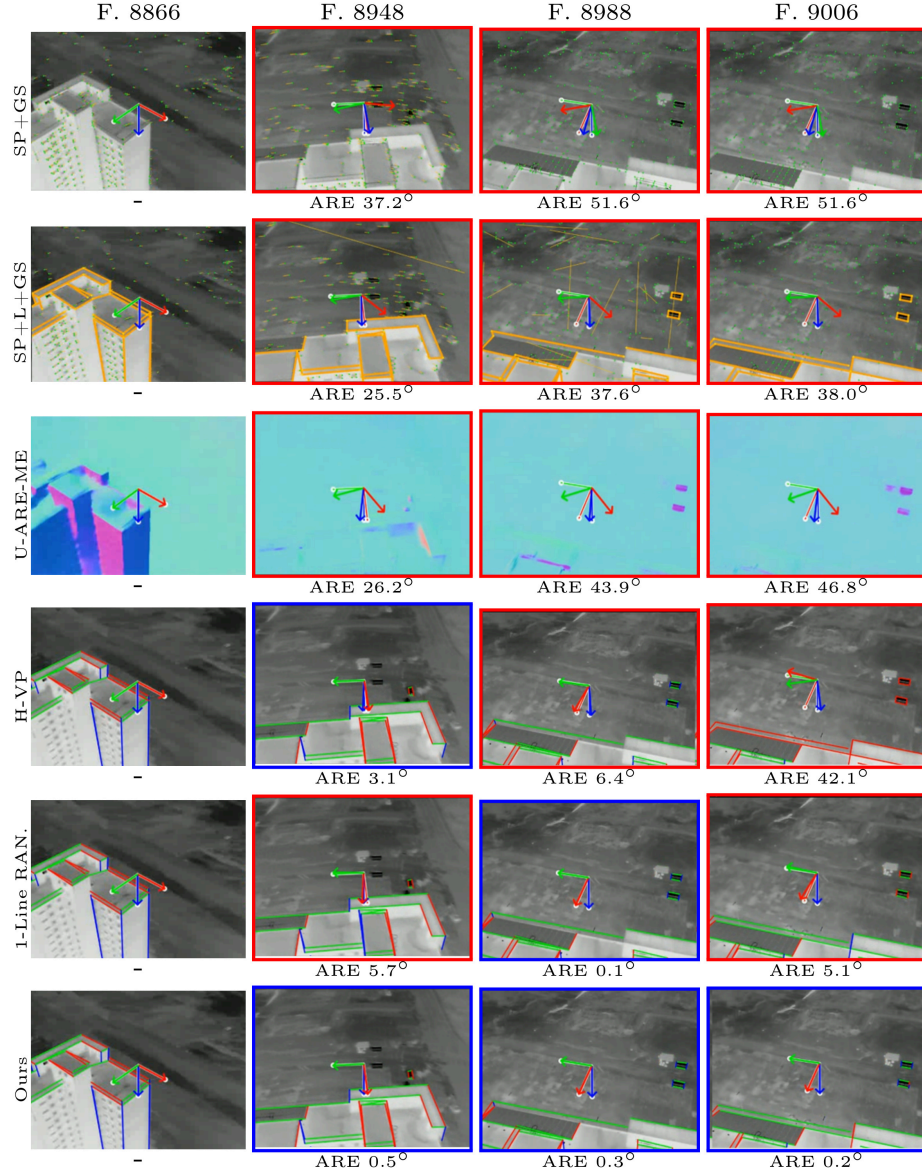


Fig. 2: Rotation estimates on the line-sparse segment of Residential-closed (frames 8866–9006), where the viewpoint sweeps past the boundary of the complex and only a few structural lines remain. Per-frame ARE is reported below each image; a dash (–) at frame 8866 marks the reference frame against which the subsequent errors are measured.

while the RTK-GPS priors keep the camera positions metrically anchored. The resulting poses provide a more stable reference than the drifting onboard orientation, and we therefore use them as pseudo-GT for ARE evaluation.

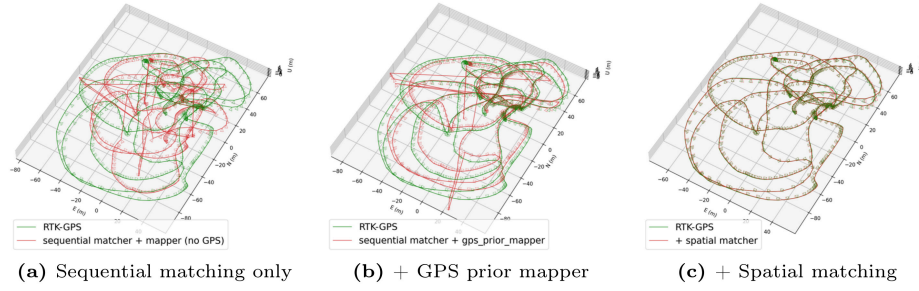


Fig. 3: Effect of RTK-GPS priors and matching strategy on the COLMAP reconstruction of Residential-closed, evaluated against the RTK-GPS positions without any post-processing. Frames are subsampled at 1 fps for this comparison.

3 Example Images and Motion Description of Our Dataset



Fig. 4: All six sequences of our benchmark. Each row shows representative RGB-thermal frame pairs. The last column depicts the camera motion: per-frame yaw profiles for the pure-rotation sequences (S-, N-, C-Buildings, top) and RTK-GPS trajectories color-coded by speed for the 6-DoF sequences (S1 Building, Residential-closed/opened, bottom).

4 Qualitative Evaluation of Monocular SLAM on Thermal Imagery

Fig. 5 compares four classes of monocular SLAM baselines, ORB-SLAM3 [1] (classical feature-based), DROID-SLAM [8] (deep dense), HI-SLAM2 [9] (geometry-aware GS), and PI-EAT-SLAM [4] (thermal-specific edge-based GS), against our pseudo-GT, complementing Tab. 4 of the main paper.

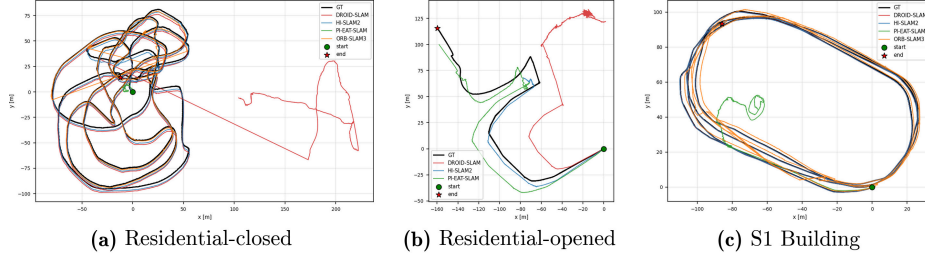


Fig. 5: Qualitative trajectory comparison of monocular SLAM baselines on the three 6-DoF sequences, evaluated against the RTK-anchored pseudo-GT (black).

On the Residential-closed sequence (fastest motion, largest area), ORB-SLAM3 and DROID-SLAM lose tracking mid-flight, whereas HI-SLAM2 completes the sequence and PI-EAT-SLAM produces a severely distorted trajectory. On the Residential-opened sequence (near-pure-rotation segments, no loop closure available), the outcome reverses: only the edge-aware PI-EAT-SLAM tracks to the end, while the others fail to initialize (ORB-SLAM3) or drift over the rotation-dominant segments (DROID-SLAM, HI-SLAM2). On the S1 Building sequence (repeated circular flight), the learning-based baselines (DROID-SLAM, HI-SLAM2) stay close to the pseudo-GT, whereas PI-EAT-SLAM fails under the faster motion. No single baseline is thus robust across all three sequences; each fails when its underlying tracking cue, such as feature matching, photometric convergence, or sufficient translational parallax, becomes unreliable. In contrast, our per-frame line-regularity module remains available as an independent orientation cue in these regimes (cf. Sec. 4.7 of the main paper).

References

1. Campos, C., Elvira, R., Rodríguez, J.J.G., M. Montiel, J.M., D. Tardós, J.: Orbslam3: An accurate open-source library for visual, visual-inertial, and multimap slam. *IEEE Transactions on Robotics* **37**(6), 1874–1890 (2021). <https://doi.org/10.1109/TR0.2021.3075644> 6
2. Chen, Q., Shu, S., Sun, H., Chen, J., Bai, X.: Thermal3d-gs: Physics-induced 3d gaussians for thermal infrared novel-view synthesis with a large-scale dataset. *IEEE Transactions on Pattern Analysis and Machine Intelligence* **48**(6), 6962–6979 (2026). <https://doi.org/10.1109/TPAMI.2026.3663966> 3

3. Delaune, J., Hewitt, R., Lytle, L., Sorice, C., Thakker, R., Matthies, L.: Thermal-inertial odometry for autonomous flight throughout the night. In: 2019 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS). pp. 1122–1128 (2019). <https://doi.org/10.1109/IROS40897.2019.8968238> 3
4. Kim, J., Yeom, S., Yang, S., Ham, J., Shin, H., Kim, C., Lee, H., Kim, P.: Pi-eat-slam: Physics-informed edge-aware thermal gaussian splatting slam. In: IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS) (2026) 6
5. Lu, R., Chen, H., Zhu, Z., Qin, Y., Lu, M., Zhang, L., Yan, C., Xue, A.: Thermal-gaussian: Thermal 3d gaussian splatting. In: The Thirteenth International Conference on Learning Representations (2025), <https://openreview.net/forum?id=ybFRoGxZjs> 3
6. Özer, M., Weiherer, M., Hundhausen, M., Egger, B.: Exploring multi-modal neural scene representations with applications on thermal imaging. In: Del Bue, A., Canton, C., Pont-Tuset, J., Tommasi, T. (eds.) Computer Vision – ECCV 2024 Workshops. pp. 82–98. Springer Nature Switzerland, Cham (2025) 3
7. Pautrat, R., Liu, S., Hruby, P., Pollefeys, M., Barath, D.: Vanishing point estimation in uncalibrated images with prior gravity direction. In: Proceedings of the IEEE/CVF International Conference on Computer Vision. pp. 14118–14127 (2023). <https://doi.org/10.1109/ICCV51070.2023.01298> 1
8. Teed, Z., Deng, J.: DROID-SLAM: Deep visual SLAM for monocular, stereo, and RGB-D cameras. In: NeurIPS (2021) 6
9. Zhang, W., Cheng, Q., Skuddis, D., Zeller, N., Cremers, D., Haala, N.: Hi-slam2: Geometry-aware gaussian slam for fast monocular scene reconstruction. IEEE Transactions on Robotics 41, 6478–6493 (2025). <https://doi.org/10.1109/TR0.2025.3626627> 6
10. Zhong, C., Xu, C., Hao, R., Zhao, H.: Tex-nerf: Neural radiance fields for novel hadar view synthesis. In: 2025 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS). pp. 9902–9908 (2025). <https://doi.org/10.1109/IROS60139.2025.11246630> 3